

Instructions: Use proper notation, and show all of your work to receive credit. Round numeric answers to at least 2 decimal places.

1. (1 pt.) Suppose that $\hat{\theta}_n \xrightarrow{P} \theta$ and $\hat{\gamma}_n \xrightarrow{P} \gamma$. Then,

$$\hat{\theta}_n + \hat{\gamma}_n \xrightarrow{P} \underline{\hspace{2cm}}.$$

Solution: Section 9.3 lecture notes, Theorem 9.2.

$$\theta + \gamma$$

2. (2 pts.) Given two unbiased estimators of θ , $\hat{\theta}_1$ and $\hat{\theta}_2$, what is the formula to find the **efficiency** of $\hat{\theta}_1$ relative to $\hat{\theta}_2$?

$$\text{eff}(\hat{\theta}_1, \hat{\theta}_2) = \underline{\hspace{2cm}}$$

Solution: Section 9.2 lecture notes, Definition.

$$\text{eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{V(\hat{\theta}_2)}{V(\hat{\theta}_1)}$$

3. (2 pts.) **Cramer-Rao Inequality:** Suppose $Y_1, \dots, Y_n$ are i.i.d. with pdf $f(y)$ (which has θ in it). If $\hat{\theta}$ is an unbiased estimator of θ , then under very general conditions,

$$\underline{\hspace{4cm}},$$

where $I(\theta) = nE \left[- \frac{d^2 \ln(f(Y))}{d\theta^2} \right].$

Solution: Section 9.2 lecture notes.

$$V(\hat{\theta}) \geq \frac{1}{I(\theta)}$$

4. (4 pts.) An organic chemist randomly sampled ten 4.85-g aniline specimens to produce acetanilide. The dry yield of acetanilide was measured for each specimen—the mean dry yield was 3.781 grams, and the standard deviation was 0.181 grams. Prior research shows that dry yields are normally distributed. Calculate a 95% confidence interval for the true mean dry yield of acetanilide.

Solution: Section 8.8.1, Example 13.

$$\begin{aligned}\bar{y} \pm t_{\alpha/2, n-1} \left(\frac{s}{\sqrt{n}} \right) &\Rightarrow 3.781 \pm t_{.025, 9} \left(\frac{.181}{\sqrt{10}} \right) \\ &\Rightarrow 3.781 \pm .1295 \\ &\Rightarrow (3.6515, 3.9105)\end{aligned}$$

5. (8 pts.) Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Poisson}(\lambda)$. Show that $\bar{Y}$ is an efficient estimator of λ .

Solution: Homework problem 9.8b.

First, $E(\bar{Y}) = \lambda$, so $\bar{Y}$ is unbiased for λ .

Next, note that $V(\bar{Y}) = \lambda/n$. Thus, if we can show that $V(\bar{Y}) = 1/I(\lambda)$, then we know that $\bar{Y}$ is an efficient estimator of λ .

The probability mass function (pmf), $p(y)$, of a $\text{Poisson}(\lambda)$ distribution is

$$p(y) = \frac{\lambda^y e^{-\lambda}}{y!}, \quad y = 0, 1, \dots$$

We then proceed in calculating $I(\lambda)$:

$$\ln[p(Y)] = Y \ln(\lambda) - \lambda - \ln(Y!)$$

$$\frac{d \ln[p(Y)]}{d\lambda} = \frac{Y}{\lambda} - 1$$

$$\frac{d^2 \ln[p(Y)]}{d\lambda^2} = -\frac{Y}{\lambda^2}$$

$$E \left[-\frac{d^2 \ln[p(Y)]}{d\lambda^2} \right] = E(Y/\lambda^2) = \frac{1}{\lambda^2} E(Y) = \frac{1}{\lambda^2} \cdot \lambda = \frac{1}{\lambda}$$

$$\frac{1}{I(\lambda)} = \frac{1}{n E \left[-\frac{d^2 \ln[p(Y)]}{d\lambda^2} \right]} = \frac{1}{n/\lambda} = \frac{\lambda}{n}.$$

Thus, $\bar{Y}$ is unbiased for λ and $V(\bar{Y}) = \frac{\lambda}{n} = 1/I(\lambda)$. Hence, $\bar{Y}$ is an efficient estimator of λ .

6. (4 pts.) Let $Y_1, Y_2, \dots, Y_n$ denote a random sample from the probability density function

$$f(y) = \begin{cases} 2/y^2, & y \geq 2 \\ 0, & \text{elsewhere} \end{cases}.$$

Does the Law of Large Numbers apply to $\bar{Y}$ in this case? Why or why not?

Solution: Homework problem 9.32.

$$E(Y^2) = \int_2^\infty y^2 \cdot \frac{2}{y^2} dy = \int_2^\infty 2 dy = \infty.$$

Since $V(Y) = E(Y^2) - [E(Y)]^2 = \infty$ as well, the Law of Large Numbers does not apply (since this law assumes a finite variance).

7. (4 pts.) If Y has a binomial distribution with n trials and success probability p , show that Y/n is a consistent estimator of p .

Solution: Homework problem 9.20.

$E(Y/n) = \frac{np}{n} = p$, so Y/n is unbiased for p .

Note that $V(Y/n) = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}$, so

$$\lim_{n \rightarrow \infty} V(Y/n) = \lim_{n \rightarrow \infty} \frac{p(1-p)}{n} = 0.$$

Hence, by Theorem 9.1, Y/n is a consistent estimator of p .